

Shokhi (■■■) — Platform & Architecture Decision

Web app vs. mobile app: how to actually reach every woman, including rural low-literacy users

The question

Should Shokhi be built as a **web app** or a **native mobile app**, given that it must serve the full spectrum of women — from urban teenagers to rural women who may not be able to read?

Decision

Build a web app (Streamlit / progressive web) with a voice-first path — not a native mobile app. The web app is the demo and the near-term product. The true channel for rural, low-literacy women is a **voice IVR phone hotline**, which the same backend is designed to power as the production roadmap.

Why not a native mobile app

- Rural low-literacy women often **do not own smartphones** or use app stores — a native app cannot reach the highest-need group. This is exactly why the existing Ananya app (Android-only) misses them.
- A native Android/iOS build is a **large time sink** for a short solo hackathon sprint, with little payoff for the target user.
- App installation itself is a barrier (storage, data, literacy to navigate the store).

Why a web app

- Runs on **any device with a browser** — a health worker's phone, a shared smartphone, a community-center laptop, an urban teen's phone — with **nothing to install**.
- **Fast to build** and **easy to host**, satisfying the hackathon's live-demo requirement.
- Supports a **voice-first path** (speak Bangla, hear Bangla back) so **literacy is not required**.
- The UI is fully decoupled from the core, so the same brain later powers a phone hotline.

The key insight: voice, and eventually a hotline

Research is blunt: rural illiterate women use **basic button phones** and **cannot read text**. The real production channel for them is an **IVR voice hotline** — dial a number, speak in Bangla, hear guidance back. No smartphone, no literacy, no app. Shokhi is designed for this from day one.

Architecture: one Gemma brain, many front doors

User	Front door	Status
Urban teen / literate woman	Web app (text + voice)	Built now — the demo
Health worker / NGO field staff	Same web app (checklist mode)	Built now
Rural, low-literacy woman	IVR voice hotline (Bangla speech in/out)	Roadmap — same backend

Because the triage engine and Gemma backend are completely separate from the UI, the exact same core powers a web app today and a phone hotline later. This is what lets Shokhi honestly claim to serve the phone-only, can't-read user — the group every existing tool leaves out.

Decision recorded for the Build with Gemma 4 hackathon project 'Shokhi'. Web app is the built demo; IVR voice hotline is the documented production path for rural reach.